

The Robertson Chemical Kinetics Problem

Stiff ODEs: how one fast process pins the global step size

Group 5 (names / student IDs to be filled in)

Numerical Analysis of ODEs and PDEs — Project 1, Topic 3

September 17, 2026

Three chemical species — how much of each is there?

A three-species reaction system (the Robertson problem)

$$\begin{aligned}y_1' &= -0.04 y_1 + 10^4 y_2 y_3 \\y_2' &= 0.04 y_1 - 10^4 y_2 y_3 - 3 \times 10^7 y_2^2 \\y_3' &= 3 \times 10^7 y_2^2 \\y(0) &= (1, 0, 0)^T, \quad t \in [0, 40]\end{aligned}$$

In one sentence

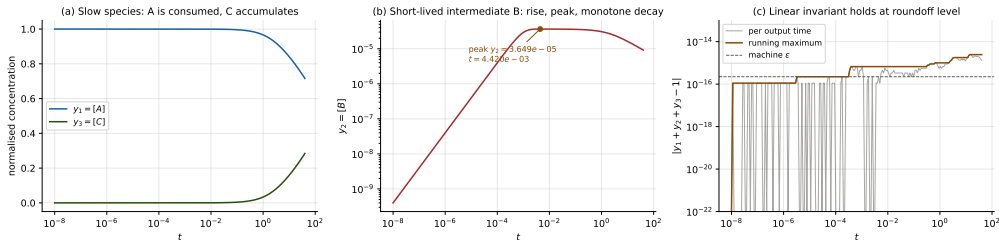
Species A, B and C interconvert, and at the start everything is A. We want to know how much of each is present at **every instant** of the 0–40 second window.

What we have to hand in

- three concentration curves $y_1(t), y_2(t), y_3(t)$
- one number: $y_2(40)$
- plus an argument: *why you should believe it*

What the answer looks like: one sharp spike, two slow curves

F2 Tight-tolerance reference (Radau, tightened tolerances)



Start with B (the middle curve)

B shoots up from zero, peaks at 3.65×10^{-5} at $t = 4.42 \times 10^{-3}$ s, then decays by 5 orders of magnitude, down to 9.18553×10^{-6} at $t = 40$ s.

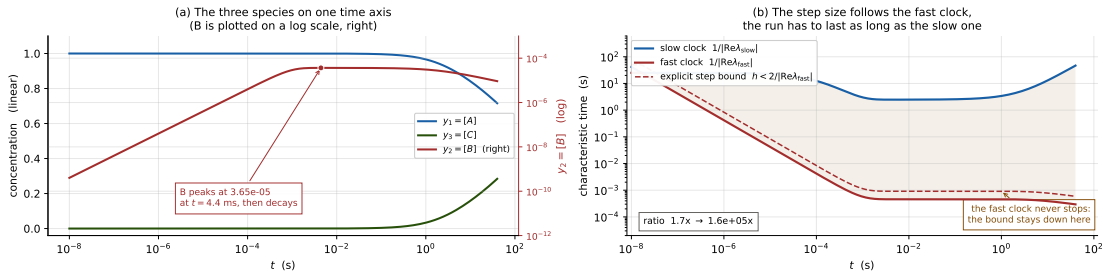
Now A and C

A drifts down from 1 to 0.7158; C drifts up from 0 to 0.2842. Both of these move on a scale of **tens of seconds** — a completely different world from B.

F2: the reference solution, with time on a logarithmic axis.

Why it is hard: B is a clock that ticks in microseconds

F13 What the system looks like, and the two clocks that make it expensive



F13: the three concentrations (left) and their clocks (right), meaning the characteristic times $1/|\text{Re } \lambda|$.

B is extremely reactive: the moment it forms it is consumed within microseconds, so its concentration never leaves the 10^{-5} range. The system therefore carries **two clocks of wildly different rate** — only $1.7\times$ apart at the start, but $1.58 \times 10^5\times$ apart by $t = 40$.

The consequence: the fastest clock pins the step size

One hard upper bound

The stability region of an explicit method only covers $[-2, 0]$ on the real axis, so the step must satisfy

$$h < \frac{2}{|\lambda|_{\max}} = 5.9 \times 10^{-4} \text{ s}$$

Crossing $[0, 40]$ therefore takes at least

$$\frac{40}{5.9 \times 10^{-4}} \approx 6.8 \times 10^4 \text{ steps.}$$

The awkward part

That $|\lambda|_{\max}$ is the maximum over the **whole interval**, and it occurs at $t = 40$, not at the start. Long after B has settled and nothing in the system is changing quickly, the restriction is *still there*.

The fast process **pins the global step size** — that is what stiffness means.

How we did it: build a ruler, then measure three methods

Step one: build a ruler

Compute a **reference solution** with a high-accuracy solver, then confirm it by two independent routes, agreeing to **10 digits**. *Without that ruler the words “how accurate” mean nothing.*

Step two: pick three methods

Explicit Euler (1) RK4 (4) Implicit Euler (1)

All three written from scratch — no off-the-shelf solver.

Step three: put every method through four checks

Check	The question it asks
conservation	does the total $y_1 + y_2 + y_3$ hold
non-negativity	any concentration negative
accuracy	pointwise error against the reference
endpoint	is $y_2(40)$ right

The four must be read separately — that is the biggest trap in this problem.

Step one: build a ruler we can trust

Three mutually independent confirmations (T4)

- **Tolerance repetition:** rerun the same solver at tighter tolerances and see at which digit the numbers stop moving
- **Cross-validation:** compare against SciPy's BDF and LSODA
- **Structural identity:** $m2(J)$ must equal $\det(B_{\text{red}})$

Confirmation route	agreeing digits
tolerance repetition	10.76
vs BDF	10.70
vs LSODA	10.98
lower bound used	10.76

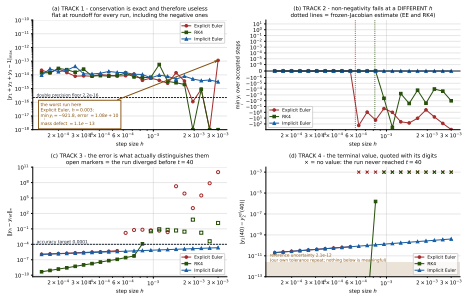
Largest componentwise deviation 2.089×10^{-12} .

This is the yardstick for everything that follows

The reference gives $y_2(40) = 9.185535 \times 10^{-6}$, quotable to 10 digits. Every error quoted later is an **pointwise comparison against it**.

Conclusion 1: “stable” is not the same as “right”

F9 Four diagnostics, four verdicts: conservation cannot stand in for accuracy



F9: conservation / non-negativity / error / endpoint, four tracks

The most glaring run (explicit Euler, $h = 3 \times 10^{-3}$)

conservation defect = 1.1×10^{-13} (looks perfect)

$\min_i y_i = -921.8$ (the concentration is negative!)

error = 1.08×10^{10}

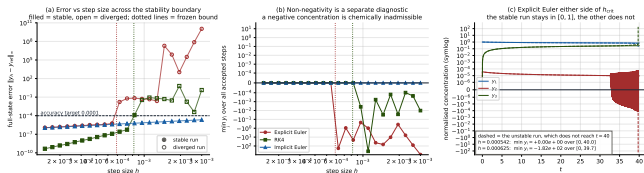
Check only conservation and this rubbish passes.

The four criteria do not agree

- the conservation defect is $\sim 10^{-14}$ at every step size — **no discriminating power at all**
- RK4 has a window that is stable, non-negative and *already inaccurate*
- the track that fails first is *not the same* for each method

Conclusion 2: the step-size limit can be predicted in advance

F7 Frozen-jacobian estimate vs an empirical step sweep



F7: 22-point logarithmic step sweep vs the frozen bound

units of 10^{-4}	EE	RK4
frozen bound	5.895	8.209
largest stable h	5.4159	8.3088
first divergent h	6.2464	9.5827

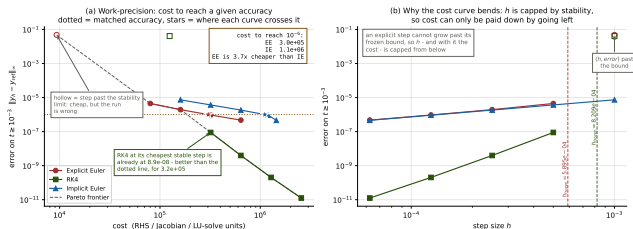
The bound falls **between two adjacent sweep points**: EE fails 8% earlier than predicted, RK4 holds 1.2% longer.

Stated honestly

This is a *locally linear*, *frozen* estimate, while $J(t)$ moves and the system is strongly non-normal. So it is a diagnostic, not a theorem.

Conclusion 3: implicit methods are not cheap on this problem

F12 Cost at matched accuracy - the implicit method is not automatically cheaper



F12: who is more expensive *at matched accuracy*

target error	EE	RK4	IE
10^{-5}	0.80	3.2	1.6
3×10^{-6}	1.1	3.2	3.9
10^{-6}	3.0	3.2	11
10^{-7}	—	3.2	—

Units of 10^5 cost units.

Three readings

- 1 IE costs **2.0–3.7** $\times$ EE at every target it can reach
- 2 at 10^{-7} **neither** EE nor IE gets there — only RK4
- 3 the cost floor is pinned by $40/h$, and h is capped by the stability bound

Conclusion 4: two practical recommendations

Rec. 1: the Newton tolerance has an **upper** bound, not just a “tight enough” one

The explicit predictor’s residual is $O(h^2)$, so a loose tolerance passes the first residual test and **Newton takes zero updates — the step is really explicit Euler**. The threshold:

$$h^* = \sqrt{\text{tol} / \|J_f f\|_\infty}$$

tolerance	error inflation	h^*
10^{-6}	2.67	1.1×10^{-2}
10^{-8}	0.88	1.1×10^{-3}
10^{-10}	1.00	1.1×10^{-4}

10^{-8} suffices; tighter is waste. But 10^{-6} pushes h^* past the stability bound 5.9×10^{-4} by $19\times$ — **and the estimator sees nothing**.

Rec. 2: adaptivity is stability-limited, not accuracy-limited

tol	$h/2 \div$ frozen bound	err/tol
10^{-4}	1.07	0.17
10^{-6}	1.07	0.19
10^{-8}	1.06	0.31

At the three looser tolerances $h/2$ sits 6–7% above the frozen bound while err/tol is only 0.17–0.31: **the accuracy budget is nowhere near spent**.

In one sentence

Where the difficulty comes from

The system contains two clocks 10^5 apart, and **the faster one pins the global step size** — the restriction does not relax even after that part has long since settled down. That is stiffness.

The methodological lesson

Verification is easier to fool yourself with than solving: conservation will fool you (defect 10^{-13} , solution wildly wrong); stability will fool you (“it did not blow up” is not “it is right”). Only a pointwise comparison against an **independently built ruler** counts.

“More stable” and “cheaper” are two different things

On this problem “implicit is more stable” holds; “implicit is cheaper” **does not**, anywhere in the reachable accuracy range. You only see that by putting cost, error and the stability bound on one plot.

Reproducibility and AI transparency

One command recomputes everything

```
python run_all.py — about 6 minutes, 10 stages  
reference / eigen / stability / integrators  
/ sweep /  
diagnostics / newtonsweep / adaptive /  
newtonstudies / figures
```

- **not one number** in results/ is typed by hand
- SciPy is used *only* as a reference to compare against; EE / RK4 / IE / Newton are **all implemented here**
- 148 tests green, including several *deliberate negative-result* regressions

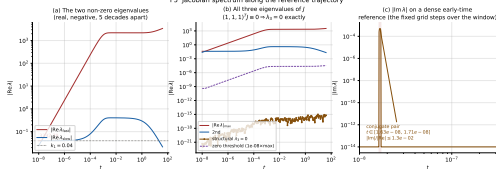
AI transparency log (18 entries; full list in appendix/ai_transparency_log.md)

(1) reduced Jacobian as a submatrix · (2) unusable significant-digit rule · (3) wrong RK4 stability polynomial · (4) “a fixed grid resolves collisions” · (5) first uniform step ate the interval · (6) inexact-Newton claim did not transfer · (7) measured bound assumed exact · (8) Newton tolerance as an unlimited knob · (9) absolute-norm controller missed the stiff mode · (10) conservation as a universal gate · (11) **a draft slide row did not match its table** · (12) **first draft ordered by computation** · (13) **the same wrong row left in the README** · (14) **three adaptive defects did not match T12** · (15) **damped-Newton tests on the wrong system** · (16) **Jacobian checker transposed — and rejected a correct matrix** · (17) **“a loose tolerance only costs accuracy” is false** · (18) **T16 measured a cold start, not the driver path**

Each entry records what was wrong, how it was found, and what it became.

Backup: the spectrum — one eigenvalue that is identically zero

F5 Jacobian spectrum along the reference trajectory



F5: the eigenvalues as functions of time

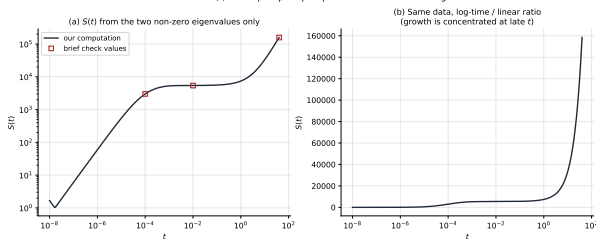
- Because $(1, 1, 1)^T$ is a **left null vector** of J , $\lambda_3 \equiv 0$ and $\text{rank } J \equiv 2$ — an exact structural property, not a numerical accident.
- **The reduced Jacobian is not a submatrix:** the correct construction is the projection $B_{\text{red}} = SJT$, not the top-left 2×2 block. Checking $m2(J) = \det(B_{\text{red}})$ gives a largest relative deviation of 1.80×10^{-15} — **pass**.
- **The complex-pair window is extremely narrow:** only 9 states have a negative discriminant, all inside $t \in [1.63, 1.71] \times 10^{-8}$, largest $|\text{Im}|/|\text{Re}| = 0.0129$; the 201-point output grid samples **none of them**.

Therefore

In practice there are always just two *real* non-zero eigenvalues.

Backup: the stiffness ratio $S(t)$ against the brief

F6 Stiffness ratio $S(t) = \max|\operatorname{Re}\lambda|/\min|\operatorname{Re}\lambda|$ over the two non-zero eigenvalues



F6: $S(t) = \max|\operatorname{Re}\lambda|/\min|\operatorname{Re}\lambda|$, counting only the two non-zero eigenvalues

t	our S	brief's value
10^{-4}	3.0097×10^3	3.0×10^3
10^{-2}	5.4221×10^3	5.4×10^3
40	1.5840×10^5	1.58×10^5

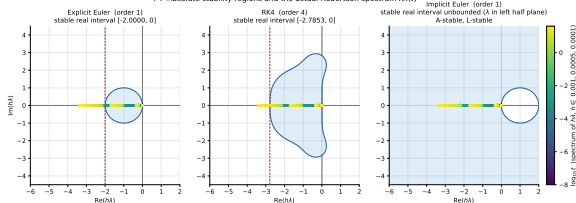
All three points agree with the brief's order-of-magnitude values to 0.25%–0.41%, and in direction: S rises monotonically to 1.58×10^5 .

The stiffest moment is $t = 40$, not $t = 0$

$|\lambda|_{\max} = 3392.79$ occurs at $t = 40$, which decides which λ the frozen bound has to use.

Backup: stability regions — bounded for explicit, unbounded for implicit

F4 Absolute stability regions and the actual Robertson spectrum $h\lambda(t)$



F4: absolute stability regions of the three methods

Method	real-axis interval	order
Explicit Euler	$[-2.0000, 0]$	1
RK4	$[-2.7853, 0]$	4
Implicit Euler	$(-\infty, 0]$	1

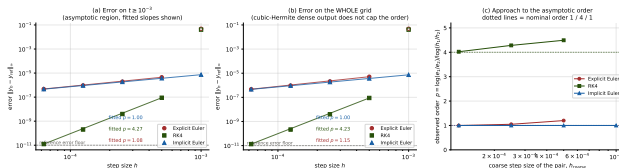
Implicit Euler is both A -stable and L -stable, so along the real axis there is **no** step-size limit; the frozen bound therefore only constrains the two explicit methods: $h_{EE} < 5.895 \times 10^{-4}$, $h_{RK4} < 8.209 \times 10^{-4}$.

Where the cost floor comes from

Substituting $|\lambda|_{\max} = 3392.79$ at $t = 40$ gives a bound that must be respected for the *whole* time span — so $[0, 40]$ needs at least 6.8×10^4 steps.

Backup: observed order of convergence

F8 Uniform-step convergence, observed order and its limits



F8: error vs h (log-log)

Method	nom.	fitted	R^2
EE	1	1.079 ± 0.033	0.9982
RK4	4	4.268 ± 0.074	0.9994
IE	1	$0.99996 \pm 1.2 \times 10^{-5}$	1.0000

Fit window $t \geq 10^{-3}$.

Why this needed its own check

Every method is densified onto the same 201-point grid by cubic Hermite interpolation *before* comparison. Were that densification only $O(h^4)$, RK4's observed order would be clipped to 2 — the measured 4.27 (4.23 over the full grid) shows it is **not**.

Backup: the window that is stable but not accurate

h	method	cons. defect	$\min_i y_i$	error	verdict
1.50×10^{-4}	all three	$1 \sim 2 \times 10^{-14}$	0.0	$4.6 \times 10^{-10} \sim 1.2 \times 10^{-6}$	all four pass
8.31×10^{-4}	RK4	6.2×10^{-15}	0.0	1.24×10^{-4}	accuracy fails
1.27×10^{-3}	EE	2.4×10^{-15}	-42.9	2.3×10^{-2}	only conservation passes

That middle row is the whole point

Stable, non-negative, conservative — and **wrong**. Any quality gate whose only criterion is the conservation defect will let it through.

The direct consequence for the report

The four diagnostics have to be four separate columns, and it has to be reported *which one fails first*. Collapsing them into a single threshold erases this window entirely.

Backup: the digits of $y_2(40)$ depend on h

h	tol	$y_2(40)$	digits
1.0×10^{-3}	10^{-8}	9.1840386×10^{-6}	3.79
1.0×10^{-3}	10^{-12}	9.1856705×10^{-6}	4.83
2.5×10^{-4}	10^{-12}	9.1855687×10^{-6}	5.43
6.25×10^{-5}	10^{-12}	9.1855388×10^{-6}	6.36

IE converges first order for this quantity, so the digits are decided entirely by h . Tightening the Newton tolerance from 10^{-8} to 10^{-12} at the same h buys only 1.04 further digits — **the remaining digits come from cutting h , not the tolerance.**

Verdict on the brief's reference value

The brief's 9.1855×10^{-6} is confirmed by our reference solution to **5.42 digits**.

$\Rightarrow 9.1855 \times 10^{-6}$ is quotable; the next digit is not

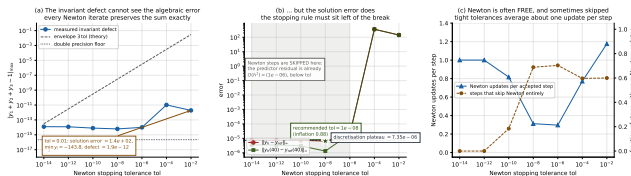
Our own $9.1855347646 \times 10^{-6}$ is quotable to 10.16 digits.

$y_2(40)$ against the y_2 peak

The peak 3.6487×10^{-5} (at $t = 4.42 \times 10^{-3}$) is **4×** the endpoint value. Quoting only the endpoint hides the transient, so we give both.

Backup: how to read the full Newton tolerance sweep

F10 Newton stopping-tolerance sweep at $h = 0.001$: the algebraic error must be negligible against the discretisation error



F10: conservation defect (flat) vs solution error (not flat)

The residual of the explicit predictor is $O(h^2)$ (about 10^{-6} at $h = 10^{-3}$), so once the tolerance reaches 10^{-6} the first residual test passes and Newton takes zero updates.

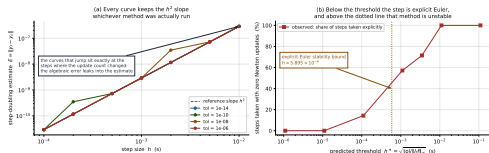
tolerance	inflation	zero-update steps
10^{-2}	1.96×10^7	60.0%
10^{-6}	2.67	70.3%
10^{-8}	0.88	68.8%
10^{-10}	1.00	18.4%

Conclusion (T11b)

The solution error plateaus at 7.3549×10^{-6} (pure discretisation error); we recommend 10^{-8} (inflation 0.88), and the first value that breaks it is 10^{-6} .

Backup: the tolerance quietly changes the method

F14 A loose Newton tolerance changes the method, not the error estimate



F14: left, the method changes while the curves do not; right, the threshold h^* against the explicit stability bound

Updates taken by the three solves (y_c , $y_{h/2}$, $y_{h/2}$)

tolerance	$h = 10^{-3}$	5×10^{-3}	10^{-2}
10^{-14}	1, 1, 1	1, 1, 1	2, 1, 1
10^{-8}	0, 0, 0	1, 1, 1	1, 1, 1
10^{-6}	0, 0, 0	0, 0, 0	0, 0, 0

0 = the predictor passes immediately, so the step is an explicit Euler step.

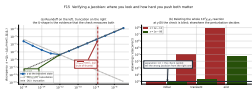
Two measurements, pointing opposite ways

- Consistent triples: the E values agree to within $1.00000 \times$, and $(0, 0, 0)$ differs from $(1, 1, 1)$ by under 1% (0.07% at $h = 10^{-3}$) — the estimator cannot tell explicit from implicit; the fitted h^2 slope stays in $[1.89, 2.10]$.
- Inconsistent triples (coarse step corrected, half-steps not): E jumps by exactly $3.00 \times$ — the algebraic error leaked into the difference.

So E reports the local error of the method *run*, not which method ran; conservation is blind too, by construction.

The threshold $h^* = \sqrt{\text{tol}/\|J_f f\|_\infty}$ was *predicted*: at $t = 1.0227$, $\|J_f f\|_\infty = 8.23 \times 10^{-3}$, so 10^{-6} gives $h^* = 1.1 \times 10^{-2}$ — 19 $\times$ the stability bound.

Backup: verifying the Jacobian independently



F15: the perturbation sweep is U-shaped; can the check see a deleted reaction term?

Protocol (T17): column-wise differences

Perturbing component j produces **column** j , not row j ; $\epsilon_j = \sqrt{\epsilon_{\text{mach}}} \max(1, |y_j|)$. $J_F = I - hJ_f$ needs its *own* tuned perturbation — it is not the same function as f .

Column-wise verification: two ways of being blind

state	η @ rule-of-thumb	η @ tuned ϵ
initial	4.47×10^{-1}	3.05×10^{-5}
transient	2.50×10^{-4}	1.94×10^{-8}
final	1.32×10^{-4}	9.25×10^{-9}

The minimum sits at $\epsilon = 10^{-12}$ and rises on both sides (cancellation left, truncation right): **report η with its state and perturbation**; there is no universal threshold.

Delete the whole $10^4 y_2 y_3$ reaction term — can the check see it?

state	separation $\epsilon = 1.5 \times 10^{-8}$	separation $\epsilon = 10^{-12}$	verdict
initial	1.0	1.0	totally blind
transient	1.9	1.2×10^4	needs the tuned ϵ
final	6.4×10^3	9.1×10^7	obvious

At the initial state $y_2 = y_3 = 0$ every entry this bug touches is already zero: **the matrices are identical**, so checking one state certifies a wrong Jacobian.

Backup: damped Newton — cold start vs driver path

What we did

The direction $J_F \delta = -F$ is computed once; the update is $x + \alpha \delta$ with α halved from 1 until the residual **strictly decreases**. If the backtrack budget or $\alpha_{\min}$ runs out first, we **report failure** and let the outer loop shrink h — rather than handing the step back.

† Our criterion is stronger than the material's:

$\|G_{\text{new}}\|^2 \leq (1 - 2c\alpha)\|G\|^2$, a rigorous consequence of $\nabla \phi^T \delta = -\|G\|^2$ for $\phi = \|G\|^2/2$.

Measured on the first step (T16)

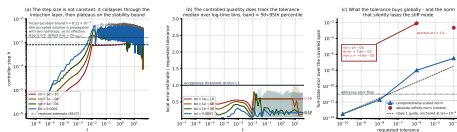
h	full	damped	backtracks	rhs
10^{-3}	5	4	1	6 $\rightarrow$ 10
10^{-1}	12	5	7	13 $\rightarrow$ 18
1	15	6	11	16 $\rightarrow$ 24
40	21	5	15	22 $\rightarrow$ 26

Both variants return the **same** state at every step (to 10^{-11}). The full step breaks the “at most 10 updates” criterion from $h = 10^{-1}$; the damped step never does — but that is **solver cost**, not accuracy.

The table is a **cold start**. On the driver path, which starts from the free explicit predictor, 4 000 steps ($h = 10^{-2}$) and 40 000 ($h = 10^{-3}$) backtrack **zero times** — while damping pays 1.50× the rhs calls for insurance that never fires. **The predictor, not the damping, makes the driver path monotone** (Entry 18).

Backup: adaptive stepping, and a failure experiment kept on purpose

F11 RK4 step-doubling controller: step size, error tracking, and a norm that fails



F11: $h(t)$, err/tol tracking, one failing norm

tol	$h/2 \div$ frozen bound	err/tol
10^{-4}	1.07	0.17
10^{-6}	1.07	0.19
10^{-8}	1.06	0.31
10^{-10}	0.83	0.59

The accepted solution advances by **two half-steps**, so $h/2$ must satisfy the stability bound: it sits 6–7% above the frozen bound.

An absolute L^∞ norm controller is broken here

$y_2 = O(10^{-5})$ but the error scale is 10^{-10} . With an absolute bound of 10^{-6} the controller accepts steps far beyond the RK4 stability bound:

- $\text{tol} = 10^{-6}$: no NaN, but $y_2 \rightarrow -4.77 \times 10^{-5}$ (**non-physical**), whole-run error 7.83×10^{-3}
- $\text{tol} = 10^{-4}$: **aborts** at $t = 1.9$, $\min y = -9.03 \times 10^2$

Both runs show a conservation defect of $\sim 10^{-15}$: they look healthy.

A component-wise weighted (mixed) norm is required:
 $\max_i |e_i| / (\text{atol}_i + |y_i|) \leq \text{tol}$; **T12** reports both.

Backup: file and figure manifest

Figures (results/figures/)

- **F2** reference solution
- **F4** stability regions
- **F5** spectrum
- **F6** stiffness ratio
- **F7** frozen bound vs step sweep
- **F8** order of convergence
- **F9** four-track diagnostics
- **F10** Newton tolerance sweep
- **F11** adaptive stepping
- **F12** work–precision
- **F13** two clocks (added this pass)
- **F14** tolerance changes the method, **F15** Jacobian check

Tables (results/tables/)

- **T1/T2** spectrum summary and brief check values
- **T3** stability regions and step limits
- **T4** reference accuracy
- **T5/T6** step sweep and the three boundaries
- **T7/T8** convergence and observed order
- **T9** cost accounting
- **T10** endpoint $y_2(40)$
- **T11/T11b** Newton tolerance sweep
- **T12** adaptive (both norms side by side)
- **T13/T14** four-track diagnostics and their boundaries
- **T15** matched-accuracy cost (added this pass)
- **T16–T18** the three Tutorial 2 studies

Backup: what this pass (P4) added

- **Rewrote the narrative** (Entry 12): the first draft followed *computation order* — every number right, still unreadable. Rebuilt as *problem* → *difficulty* → *method* → *conclusions*. No numerical check catches this class of defect.
- **F13 two_clocks**: drawn for the “why is it hard” frame. Left panel: all three species on one time axis (B is a spike). Right panel: the clocks $1/|\operatorname{Re} \lambda|$ and the explicit step limit $2/|\operatorname{Re} \lambda_{\text{fast}}|$ — stiffness stated *before* any method appears.
- **F12 work_precision**: from the cost counts in **T9**, it answers “who is cheaper *at the same accuracy*”. Left panel: work-precision and the Pareto front. Right panel: why the cost curve folds at the stability bound.
- **T15 matched_accuracy_cost**: the same question as a table, with a status column separating interpolated / already_better_than_target / not_reached_in_sweep. A missing row must state its reason, else “no EE at 10^{-7} ” reads as “not measured”.
- **Filled in the implicit side** (Entries 15–18): backtracking **damping** in `newton.py`; `jacobian_check.py` and `newton_estimator.py` added; **T16–T18, F14–F15**.

Wording discipline around “who is cheaper”

At exactly 10^{-6} error EE uses the fewest cost units (3.0×10^5 vs RK4’s 3.2×10^5); but RK4 buys 8.9×10^{-8} for the same 3.2×10^5 — $11\times$ more accuracy. Both are true and answer different questions, so the report keeps them apart.

Backup: what Tutorial 2 filled in

Three questions the material keeps apart, and the previous version merged

- 1 **Is the stepping stable** — $R(z)$ on the spectrum of J ; A- and L-stability;
- 2 **Does Newton converge** — a property of the *solve*, not of the trajectory;
- 3 **Is the discretisation accurate** — $O(h^p)$, what the estimator should track.

The previous version had 1 and 3 and treated 2 as a knob inside 3; **two of the three then changed a claim, not just a number.**

The sentence that matters most

A successful solve says nothing about the accuracy of the step it produced. In **T16** the same root is reached to 10^{-11} at very different costs; in **T18** a “solve taking zero updates” turns implicit Euler into explicit Euler.

The three deliverables

- **T16** + **newton.py**: on a cold start the full-step direction count goes $3 \rightarrow 21$ while damping stays in $[3, 6]$, both returning the **same state** (10^{-11}); on the **driver path damping never fires** (Entry 18).
- **T17** + **F15**: a *correct* matrix already scores $\eta = 0.447$ under the rule-of-thumb perturbation; delete a reaction term and the initial-state separation is **exactly 1.000** — **one state is not a check.**
- **T18** + **F14**: $h^* = \sqrt{\text{tol}/\|J_{ff}\|_\infty}$ matches the observed explicit steps 7/7, while E moves by under 1%.
- Adaptive controller: **a failed Newton solve must reject the step**; rejected steps are excluded from the statistics.